Workshop on EV Battery and Charging Technology

Organized under: IEEE IES HUBS & NODES Initiative – Hyderabad Hub, India
Dates: January 5–6, 2026
Venue: Room No. 203, NAC–2, IIT Madras, India
Organized By: Department of Engineering Design, IIT Madras

Overview

The two-day workshop focused on Electric Vehicle (EV) battery systems, charging infrastructure, wireless charging, and Vehicle-to-Grid (V2G) technologies. The event brought together students, researchers, faculty members, and industry professionals to explore advancements in sustainable mobility and industrial electronics. The workshop marked the first officially conducted IEEE IES HUBS & NODES activity in the Chennai region and promoted collaboration between academia and industry.

Participation Summary

Category	Count
UG Students	11
PG Students	20
PhD Scholars	30
Faculty Members	4
Industry Professionals	24
Total Participants	89

Technical Sessions

Session 1 – EV Battery Technology
Battery chemistry, thermal management, battery ageing, Battery Management Systems (BMS), and sustainability aspects of EV batteries were discussed.

Session 2 – Multiport Chargers for Electric Vehicles
The session focused on power electronics, power factor correction, and scalable charging systems for EVs.

Session 3 – Reconfigurable EV Charging Hubs
Topics included flexible charging infrastructure, multi-output charging systems, and efficient charging hub architectures.

Session 4 – Heavy-Duty Vehicle Charging
Megawatt charging systems, battery swapping, and charging solutions for buses, trucks, and industrial vehicles were presented.

Session 5 – Wireless Charging Technology
Wireless charging architectures, alignment challenges, and modular charging approaches were discussed.

Session 6 – Vehicle-to-Grid (V2G) Technology
The session highlighted bidirectional charging, grid integration, and energy management using EV fleets.

IEEE IES Engagement and Outcomes

IEEE IES Virtual Engagement
Participants interacted with IEEE IES leadership, including Dr. Juan José Rodríguez-Andina and Dr. Milos Manic, who shared insights into IEEE IES initiatives, global networking, and professional development opportunities.

Laboratory and Research Park Visits
Participants visited multiple laboratories at IIT Madras including Battery Charging Technology Lab, Automotive Systems Lab, and Motors & Control Lab. The visit to IIT Madras Research Park provided exposure to startups and EV innovation ecosystems.

Industry Participation
Industry professionals from organisations such as TCS, Tata Elxsi, Applied Materials, Stellantis, Magna, Atlas Copco, and Wabtec participated in the workshop and contributed industrial perspectives to EV charging and battery technologies.

Workshop Outcomes
- Enhanced awareness of EV battery and charging technologies
- Strong industry–academia interaction
- Increased awareness of IEEE IES activities
- Interest in future IEEE IES initiatives and collaborations

Conclusion
The workshop successfully delivered advanced knowledge on EV technologies while strengthening collaboration between academia, industry, and IEEE IES. The event established a strong foundation for future IEEE IES HUBS & NODES activities in the Chennai region.

Report By:
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